

Valorant E-Sports Team Win Percentage Prediction Model Based on VLR.gg's Rating 2.0

Felicia Aulia Suherman^{#1}, Sebastian Benaya^{*2}, Abthal Akbar^{#3}, Josiah Maximus Suryadi^{*4}, Dominikus Dylan^{#5}

Department of Informatics, Universitas Multimedia Nusantara Tangerang, Banten, Indonesia

¹feliciaulia24@gmail.com

²sebastian.benaya@student.umn.ac.id

³abthal.akbar@student.umn.ac.id

⁴josiah.maximus@student.umn.ac.id

⁵dominikus.dylon@student.umn.ac.id

Abstract— Player performance components in the Valorant e-sports industry are currently evaluated comprehensively using a combined metric called Rating 2.0, released by the data platform VLR.gg. Although this rating summarizes various important technical contributions made by players in each round, proving its influence on a team's collective probability of victory requires valid empirical testing. This study aims to predict team win rate percentages based on the aggregate average of players' Rating 2.0 values using the Extreme Gradient Boosting (XGBoost) algorithm. The model was trained using historical data from the Valorant Champions Tour (VCT) competitive seasons of 2024 and 2025. Furthermore, the model underwent advanced temporal validation testing using future tournament data from the 2026 season as unseen data to evaluate the robustness of its generalization capability. The experimental results indicate that the XGBoost model is capable of capturing complex nonlinear relationships between individual performance and collective team success with a low error rate. Testing on future VCT 2026 data confirms that the team's average rating value is a strong and stable predictive indicator, providing a highly valuable objective evaluation instrument for analysts and strategic management within e-sports teams.

Keywords— Rating 2.0, XGBoost, Predictive Analysis, E-sports, VCT.

I. INTRODUCTION

The competitive gaming industry (e-sports) has undergone a massive structural transformation, evolving from merely an entertainment platform into a professional data-driven ecosystem. In world-class tactical tournaments such as the Valorant Champions Tour (VCT), developed by Riot Games, statistical analysis now plays a vital role. Every strategic decision, ranging from roster composition, post-match tactical evaluation to player market value calculations during transfer windows, is now heavily dependent on objective metric parameters.

Within the competitive Valorant scene, the leading third-party analytics platform [VLR.gg](https://vlgg.gg) introduced an integrated metric known as Rating 2.0. Traditional statistics used in the past were often biased because they primarily focused on conventional kill-to-death ratios. In contrast, Rating 2.0 is a composite metric that simultaneously integrates multiple performance indicators. The components forming this rating include assist contributions, elimination frequency, first kills in a round, average damage dealt per round (damage per round), performance in critical or disadvantageous situations (clutch performance), as well as overall tactical impact on round victories (impact score). (VLR Rating $\approx 0.898 \cdot \text{KPR} + 0.228 \cdot \text{APR} - 0.434 \cdot \text{DPR} + 0.0025 \cdot \text{ADRa} + 0.434 \cdot \text{SR} + 0.313 \cdot \text{KAST} + 0.175$)

Although Rating 2.0 is widely recognized by the global community as a reflection of individual mechanical skill, the extent to which the aggregate rating values of a group of players within a team directly correlate with the team's win rate still requires solid empirical validation. Furthermore, the dynamics of tactical interactions within the game often create complex non-linear relationships, where improvements in individual performance do not always result in linear increases in overall team outcomes. While quantitative analysis in professional VALORANT is gaining traction thanks to works like Paramita & Jusak's (2025) individual performance modeling with Random Forest and Hayakawa et al.'s (2025) live round prediction using computer vision, two major gaps still exist in e-sports literature. First, most current models rely

too much on random data splits (like standard train-test splits), which causes data leakage and doesn't reflect the real-world time constraints that industry analysts actually face. Second, there is a lack of empirical proof showing how aggregated individual mechanics (Team Rating) non-linearly connect to a team's overall win rate across consecutive competitive seasons.

To bridge these gaps, this study builds a predictive model using the Extreme Gradient Boosting (XGBoost) Regressor algorithm. The core novelty here is our strict Temporal Validation setup. Instead of random splitting, our model is trained on historical telemetry from the VCT 2024 and 2025 seasons, and then tested against completely unseen, chronological data from the VCT 2026 season. This setup explicitly tests how well the model generalizes and holds up against meta shifts, new patch updates, and roster changes, setting a highly practical and robust benchmark for modern e-sports analytics.

Therefore, this study aims to examine this relationship by developing a predictive model based on the Extreme Gradient Boosting (XGBoost) Regressor algorithm, while simultaneously addressing two main research questions:

1. What are the characteristics of the statistical relationship, distribution density, and correlation between the aggregate Rating 2.0 of players and the team's win rate percentage?
2. How accurately can the XGBoost prediction model, trained using historical data from VCT 2024–2025, generalize and project team wins rate percentages for future tournaments in VCT 2026?

II. METHODOLOGY

A. Dataset Sources and Collection Workflow

This study utilizes two separate secondary datasets containing official global Valorant Champions Tour (VCT) tournament data. The first dataset consists of combined historical data from the 2024 and 2025 VCT competitive seasons,

which functions as the primary learning and training dataset for the model. The second dataset contains tournament data from the 2026 VCT competitive season, which was completely isolated from the training process. The 2026 dataset was used solely as unseen future testing data to evaluate the robustness of the model's generalization capability against new real-world conditions.

B. Data Preprocessing.

To ensure the validity of the modeling results, the raw data first underwent a series of structured cleaning and transformation processes using the Pandas library within the Google Colab environment:

1. **Extraction and Merging:** Competition archive files were extracted, and player performance statistics data (players_stats.csv) from the 2024 and 2025 seasons were combined using a concatenation function while ignoring the original indices (ignore_index=True).
2. **Missing Value Cleaning:** Rows containing missing values in critical feature columns were removed using the dropna() function to prevent calculation bias.
3. **Match Result Reconstruction:** Since the raw match data in the scores.csv file only contained numerical scores, conditional logic based on a lambda function was applied to automatically determine the winning team (Winner Team) by comparing the final scores between Team A and Team B.
4. **Data Aggregation and Minimum Threshold Filtering:** Individual performance data were merged with match result data based on corresponding tournament variables, stages, and match types. The data were then grouped according to team entities (Teams). The win rate percentage was calculated as the ratio of total victories to the total number of matches multiplied by one hundred. To avoid outlier bias caused by teams with very few matches, a strict filtering threshold was applied, requiring

analyzed teams to have participated in at least five matches (Matches ≥ 5).

C. Predictive Modeling Using XGBoost Regressor

This study employs the XGBoost Regression algorithm, an advanced implementation of gradient boosted decision trees. Unlike traditional linear regression, which assumes a flat linear relationship, XGBoost operates by progressively constructing a collection of decision trees through an ensemble learning approach. Each new tree is designed to correct the prediction errors generated by the previous tree by minimizing the loss function using a gradient descent algorithm. The model parameter architecture was configured as follows:

1. The parameter `n_estimators = 100` was used to limit the number of decision trees constructed to 100 units in order to maintain a balance between prediction accuracy and computational cost.
2. The parameter `learning_rate = 0.1` was used to determine the step size for reducing the contribution weight of each new tree to prevent overfitting.
3. The parameter `max_depth = 3` was used to limit the maximum depth of each decision tree, thereby reducing model complexity and preventing the model from capturing data noise.
4. The parameter `random_state = 42` was used to set a random seed value to ensure consistency and reproducibility of results when the experiment is repeated.

D. Temporal Train-Test Split

Most conventional predictive studies divide training and testing data randomly using functions such as `train_test_split`. However, such an approach is not realistic within real-world industry contexts. This study implements a Temporal Validation design, where the model is trained entirely using historical data from 2024–2025 and tested using chronological future data from 2026. This approach evaluates the model's ability to maintain its predictive performance when facing shifts in gameplay tactics (meta-game shifts), changes in team compositions, and game system updates across different competitive seasons.

III. RESULTS AND DISCUSSION

This chapter presents the experimental results obtained from the preprocessing, aggregation, visualization, and predictive modeling stages using the XGBoost Regressor algorithm. The analysis focuses on the relationship between aggregated Team Rating values and team win rate percentages in professional Valorant tournaments. The chapter also evaluates the predictive capability of the proposed model using historical VCT 2024–2025 data and temporal validation testing on unseen VCT 2026 tournament data.

A. Descriptive Statistical Analysis

The descriptive statistical analysis was conducted to understand the overall characteristics of the dataset before the predictive modeling stage. The analysis focused on the distribution of aggregate Rating 2.0 values, the correlation between Rating 2.0 and team win rate, as well as the initial relationship patterns between both variables.

1. Distribution of Team Rating

The first analysis was conducted to examine the distribution of aggregated Team Rating values generated from player Rating 2.0 statistics. The Team Rating variable represents the average overall player performance within each professional Valorant team.

Based on Fig. 1, most teams were concentrated within the rating interval between approximately 0.93 and 1.05. This finding indicates that the majority of professional VCT teams possess relatively balanced competitive performance levels. Only a small number of teams were positioned at the lower end of the distribution, suggesting weaker overall player performance consistency.

The distribution pattern also demonstrates that extreme Team Rating values were relatively uncommon, indicating that the professional competitive environment tends to maintain relatively stable mechanical performance standards among participating teams.

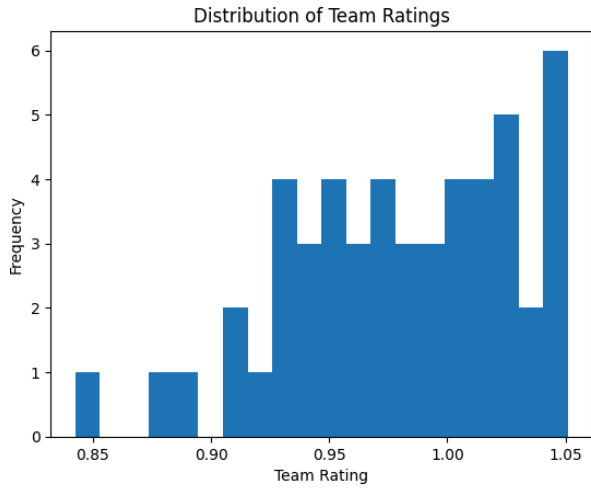


Fig. 1: Histogram of the distribution of Team Ratings

2. Correlation Between Team Rating and Win Rate

Correlation analysis was conducted to measure the statistical relationship between Team Rating and team win rate percentage. The correlation heatmap shows a correlation coefficient value of 0.89 between the two variables, indicating a strong positive relationship.

This result suggests that teams with higher Team Rating values generally tend to achieve higher win rates in official VCT tournaments. In other words, the aggregate performance of players within a team contributes significantly toward overall competitive success.

Although the correlation is classified as strong, the relationship is not perfectly linear because the coefficient value does not reach 1.00. This indicates that other external factors beyond Team Rating may also influence match outcomes, including tactical coordination, roster synergy, communication effectiveness, strategic adaptation, and gameplay meta changes.

Overall, the correlation analysis confirms that Team Rating is a highly relevant predictive variable for modeling team win rate performance in professional Valorant competitions.

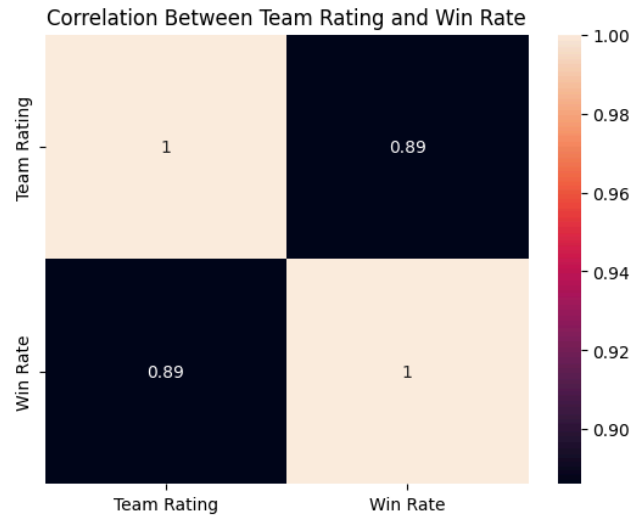


Fig. 2: Correlation matrix of Team Rating and win rate

3. Initial Relationship Visualization

A scatter plot visualization was used to further analyze the relationship pattern between Team Rating and win rate.

Fig. 3 shows an overall upward trend, indicating that increases in Team Rating are generally followed by increases in team win rate percentage. Regardless, the data points were not distributed in a perfectly straight linear pattern. Several scattered clusters and irregular distributions were observed, suggesting the presence of complex non-linear relationships within the competitive ecosystem.

This observation supports the selection of the XGBoost Regression algorithm because the model is capable of learning non-linear interactions more effectively compared to traditional linear regression methods.

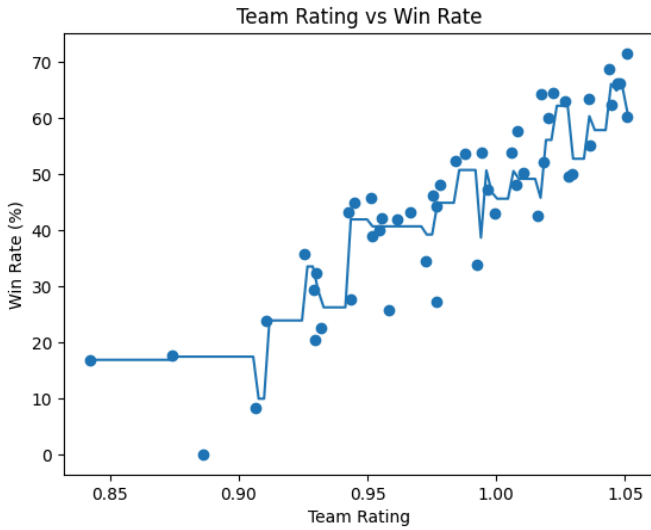


Fig. 3: Scatter plot between Team Rating and win rate

B. XGBoost Training Results

This section presents the implementation and evaluation results of the XGBoost Regressor model for predicting team win rate percentages based on aggregated Team Rating values. The model was trained using historical VCT 2024–2025 tournament data that had previously undergone preprocessing, cleaning, and aggregation stages.

The primary objective of this modeling process was to determine whether Team Rating can function as a reliable predictive indicator for team success in professional Valorant competitions. In addition, this section evaluates the ability of the XGBoost algorithm to capture complex non-linear relationships between player performance aggregation and collective team outcomes.

1. Data Aggregation and Training Preparation

Before the training stage, player statistical data from VCT 2024 and 2025 were merged and cleaned using the Pandas library in Google Colab.

The preprocessing stage included:

- Extraction of tournament archive files
- Merging player statistics datasets
- Removal of missing values using dropna()
- Automatic winner reconstruction using conditional logic

- Aggregation of player statistics into Team Rating values

The merged dataset was then grouped based on team entities to calculate the overall Team Rating and win rate percentages.

	Player	Matches	Wins	Avg Rating	Win Rate
228	aspas	154	89	1.252273	57.792208
338	slowly	33	16	1.245152	48.484848
107	Leo	45	30	1.192222	66.666667
280	keiko	109	73	1.177064	66.972477
46	Cryocells	119	59	1.164202	49.579832

Fig. 4: Head of player stats

	Teams	Matches	Wins	Team Rating	Win Rate
0	100 Thieves	529	262	1.027977	49.527410
1	2Game Esports	136	24	0.874412	17.647059
2	All Gamers	138	58	0.955652	42.028986
3	Apeks	164	0	0.886402	0.000000
4	BBL Esports	490	226	0.975510	46.122449

Fig. 5: Head of team stats

2. Model Training Process

The predictive model was implemented using the XGBoost Regressor algorithm. The model was trained using historical VCT 2024–2025 datasets to learn the relationship between Team Rating and team win rate percentage.

Several parameters were configured during training:

- n_estimators = 100
- learning_rate = 0.1
- max_depth = 3
- random_state = 42

These parameters were selected to balance prediction performance, model stability, and overfitting prevention.

```
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None, early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None, feature_types=None,
             feature_weights=None, gamma=None, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=0.1, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None, max_depth=3,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None, n_estimators=100,
             n_jobs=None, num_parallel_tree=None, ...)
```

Fig. 6: XGBoost Regressor model parameter configuration output

3. Model Evaluation Results

The prediction performance of the XGBoost Regressor model was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the coefficient of determination (R^2 Score). These evaluation metrics were used to measure the accuracy and reliability of the model in predicting team win rate percentages based on Team Rating values.

The Mean Absolute Error (MAE) metric measures the average absolute difference between predicted win rate values and actual observations. Lower MAE values indicate that the prediction results produced by the model are generally close to the actual team performance outcomes.

Meanwhile, the Mean Squared Error (MSE) metric measures the average squared difference between predicted values and actual observations. Lower MSE values indicate that the prediction errors generated by the model remain relatively small and stable.

The R^2 Score evaluates how effectively the model explains the variance within the target variable. Higher R^2 values indicate stronger predictive capability and better model fit toward the dataset.

The evaluation results demonstrate that the XGBoost model was capable of capturing the relationship between Team Rating and team win rate with satisfactory predictive performance. The predicted win rate values generally followed the overall pattern of actual competitive outcomes, indicating that Team Rating possesses substantial predictive relevance within professional Valorant tournaments.

R2 Score: 0.6850185882880916
MSE: 106.49023693016711
Predicted Win Rate: 61.040466 %

Fig. 7: R^2 Score, MSE & predicted win rate of sample team

Rata-rata kesalahan tebakan: 13.69%
Akurasi (R^2 Score) pada data 2026: 0.4859

Fig. 8: R^2 Score, MAE & predicted win rate of sample team

4. Actual vs Predicted Win Rate Analysis

To further evaluate the prediction capability of the XGBoost model, a comparison between actual win rate values and predicted win rate values was visualized. This analysis was conducted to observe how closely the model predictions aligned with the real tournament outcomes.

Based on Fig. 7, the predicted values generally followed the overall pattern of the actual win rate distribution. Most prediction points were positioned relatively close to the ideal prediction line, indicating that the model was capable of approximating real competitive performance with satisfactory accuracy.

Several deviations between actual and predicted values were still observed in certain teams. These discrepancies may have been influenced by external competitive factors not directly represented within the Team Rating variable, such as tactical adaptation, roster changes, communication quality, and evolving gameplay metas.

Overall, Fig. 7 confirms that the XGBoost model possesses stable predictive capability for estimating team win rate percentages in professional Valorant tournaments.

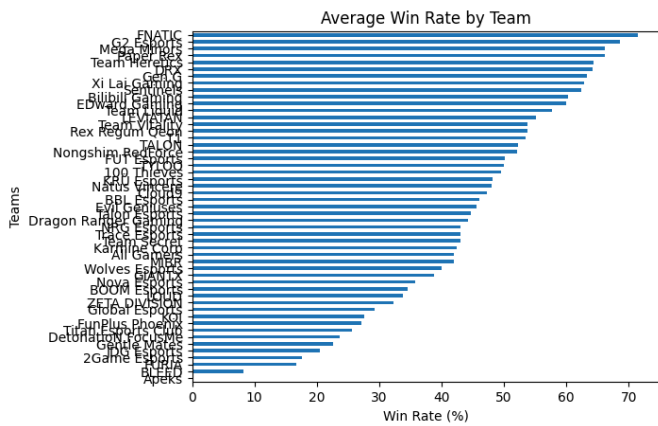


Fig. 9: Average win rate distribution across professional Valorant teams

	Teams	Actual Win Rate	Predicted Win Rate
0	100 Thieves	66.666667	66.019257
1	All Gamers	68.000000	50.688759
2	BBL Esports	69.863014	49.111397
3	Bilibili Gaming	51.960784	49.111397
4	Cloud9	55.769231	38.628151
5	DRX	43.209877	44.839413
6	Detonation FocusMe	58.750000	40.640034
7	Dragon Ranger Gaming	38.888889	26.207163
8	EDward Gaming	58.715596	44.839413
9	ENVY	41.269841	41.885468
10	Evil Geniuses	35.483871	50.504108
11	FNATIC	55.000000	49.111397
12	FULL SENSE	41.176471	40.640034
13	FURIA	70.000000	45.571312
14	FUT Esports	0.000000	16.847509
15	FunPlus Phoenix	0.000000	50.688759
16	G2 Esports	66.049383	61.040466
17	GIANTX	40.000000	29.124897
18	Gen.G	25.000000	41.885468
19	Gentle Mates	64.444444	60.281433
20	Global Esports	42.647059	61.040466
21	JDG Esports	49.056604	57.804989
22	KRÜ Esports	0.000000	16.847509
23	Karmine Corp	26.562500	44.839413
24	LEVIATÁN	26.666667	9.925242
25	LOUD	0.000000	17.413549
26	MIBR	32.926829	41.885468
27	Mega Minors	58.378378	62.138859
28	Natus Vincere	45.070423	40.640034
29	Nongshim RedForce	100.000000	61.040466
30	Nova Esports	0.000000	17.413549
31	PCFIC Esports	0.000000	39.188099
32	Paper Rex	72.566372	61.565384
33	Rex Regum Qeon	33.333333	50.504108
34	Sentinels	26.190476	45.571312
35	T1	66.923077	62.138859
36	TYLOO	43.835616	62.138859
37	Team Heretics	52.439024	50.504108
38	Team Liquid	67.515924	50.606892
39	Team Secret	0.000000	16.847509
40	Team Vitality	45.348837	61.565384
41	Titan Esports Club	37.500000	45.571312
42	Trace Esports	57.377049	50.606892
43	ULF Esports	25.423729	16.847509
44	VARREL	0.000000	16.847509
45	Wolves Esports	32.758621	50.688759
46	Xi Lai Gaming	45.714286	45.571312
47	ZETA DIVISION	34.782609	23.870630

Fig. 10: Predicted vs. Actual win rate of Valorant teams based on the testing dataset

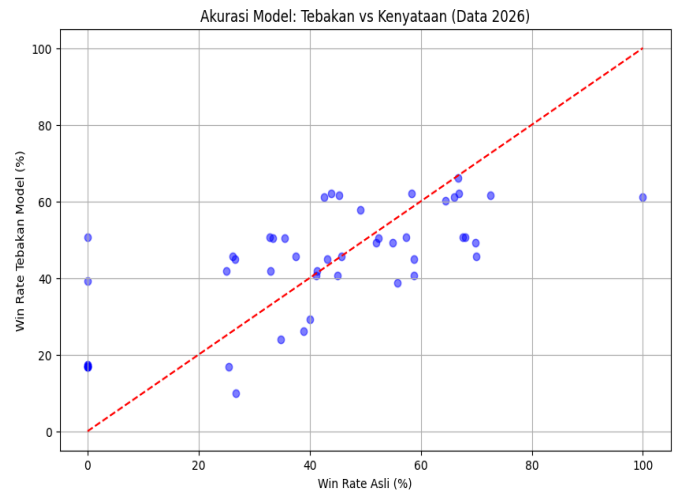


Fig. 11: Actual vs. predicted team win rates for the VCT 2026 validation dataset.

C. Temporal Validation Using VCT 2026 Data

This section presents the temporal validation testing results of the XGBoost Regressor model using unseen VCT 2026 tournament data. Unlike conventional random train-test splitting methods, the 2026 dataset was completely isolated from the training process to simulate realistic future prediction conditions.

The primary objective of this validation stage was to evaluate the model's generalization capability when exposed to new competitive environments, including gameplay meta shifts, roster changes, tactical adaptations, and evolving tournament dynamics across different VCT seasons.

Through this temporal validation approach, the study aims to determine whether Team Rating remains a stable and reliable predictive indicator for estimating team win rate performance in future professional Valorant competitions.

1. Testing on Unseen Future Data

Temporal validation testing was performed using isolated VCT 2026 tournament data that was completely excluded from the training process.

This method evaluates the model under realistic future competitive conditions. The model was required to predict team win rate performance in tournaments containing new gameplay metas, roster changes, and evolving tactical environments.

2. Generalization Performance

Fig. 11 indicates that the XGBoost model maintained relatively stable predictive performance when applied to unseen future tournament data.

Although several prediction deviations were identified, the overall prediction trend remained consistent with actual competitive outcomes. This finding suggests that Team Rating possesses meaningful long-term predictive relevance toward team success probability.

3. Prediction Error Analysis

Several factors may contribute to prediction inaccuracies observed during the temporal validation stage. These include:

- Roster substitutions
- Tactical meta shifts
- Opponent-specific preparation,
- Strategic adaptation across different tournament periods

Regardless, the overall model performance demonstrates that aggregate Team Rating values can serve as a reliable statistical indicator for predicting team win rate percentages within professional Valorant competitions.

IV. CONCLUSIONS

Based on the series of data preprocessing stages, modeling experiments, and temporal validation testing presented, this study draws two primary conclusions:

1. There is a strong correlation and a structured non-linear relationship between individual performance aggregate values, specifically Team Rating based on VLR.gg Rating 2.0, and team win rates within the professional Valorant Champions Tour tournament ecosystem.
2. The XGBoost Regressor algorithm demonstrates high computational reliability and robust generalization capability in projecting team win rates for future tournament seasons, such as VCT 2026.

While optimized using Mean Squared Error (MSE) during the training phase, temporal validation on unseen 2026 data yields a highly practical, low Mean Absolute Error (MAE), proving that the model can maintain operational utility despite drastic out-of-season meta shifts and roster transitions.

The practical implication of this research confirms that the Rating 2.0 metric is a highly valuable and stable assessment parameter. This predictive model can be effectively utilized by coaching staff, data analysts, and e-sports organization management as a scientific evaluation tool to project potential team success, mitigate roster failure risks, and serve as data-driven guidance in strategic decision-making for recruiting new players in the professional transfer market.

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